Practical Pathology

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Lab. Outline:

* Biopsy:
  - Definition
  - Types

* How to send specimen for histopathological examination?

* Tissue Processing

Biopsy:

is a piece of tissue or organ removed surgically from a patient for histopathological examination.

Autopsy

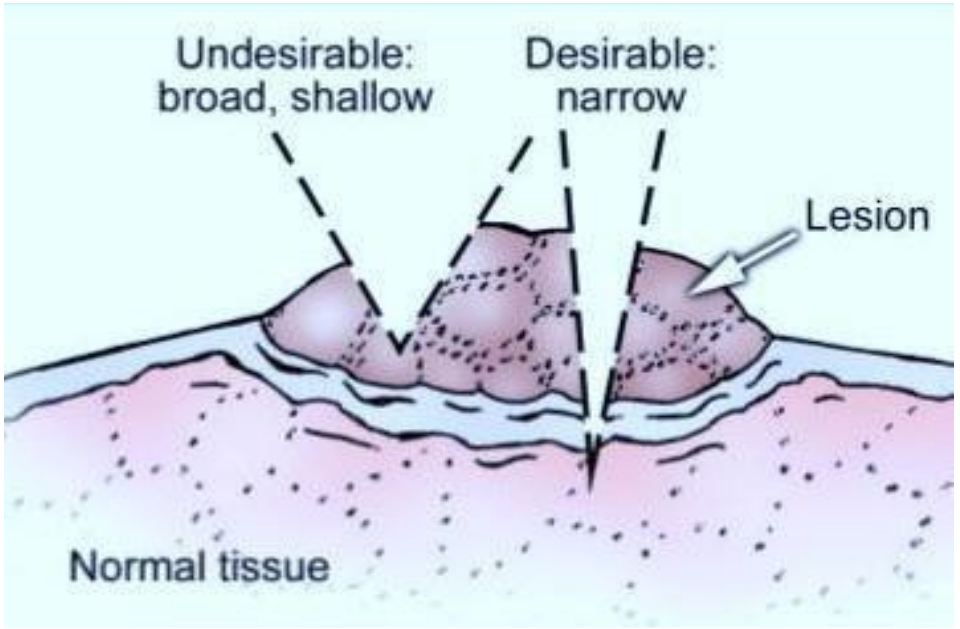
is a piece of tissue or organ taken from a dead human being for histopathological examination.

Method of Biopsy Taking:

1. Incisional biopsy
2. Excisional biopsy
3. Punch biopsy
4. Core needle biopsy
5. Curettage biopsy
6. Endoscopic Biopsy

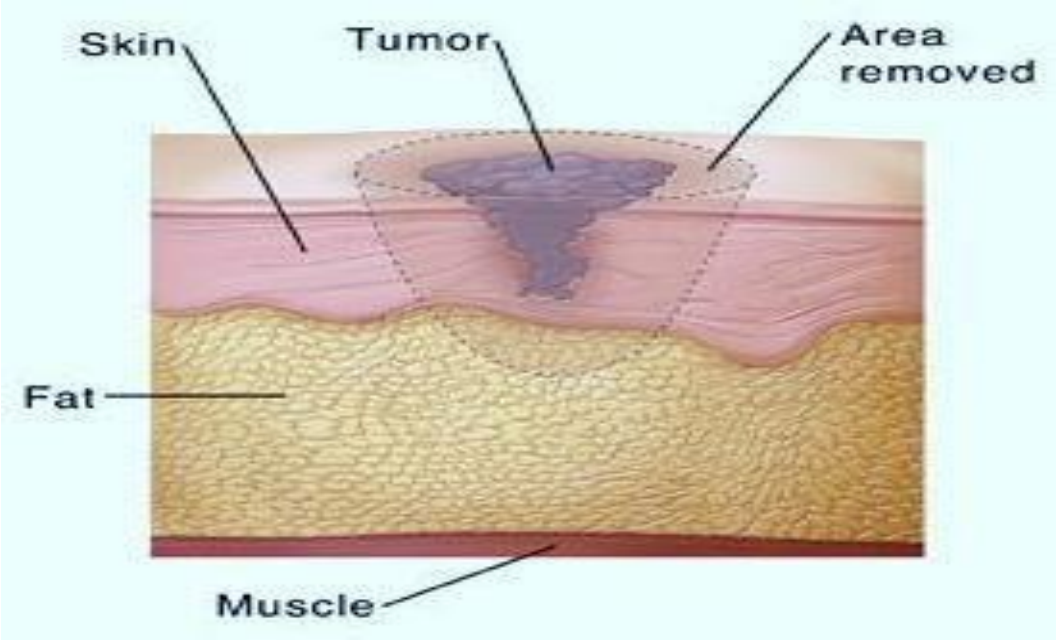
1. Incisional biopsy:

means removal of part of diseased tissue for histopathological examination.



2. Excisional biopsy:

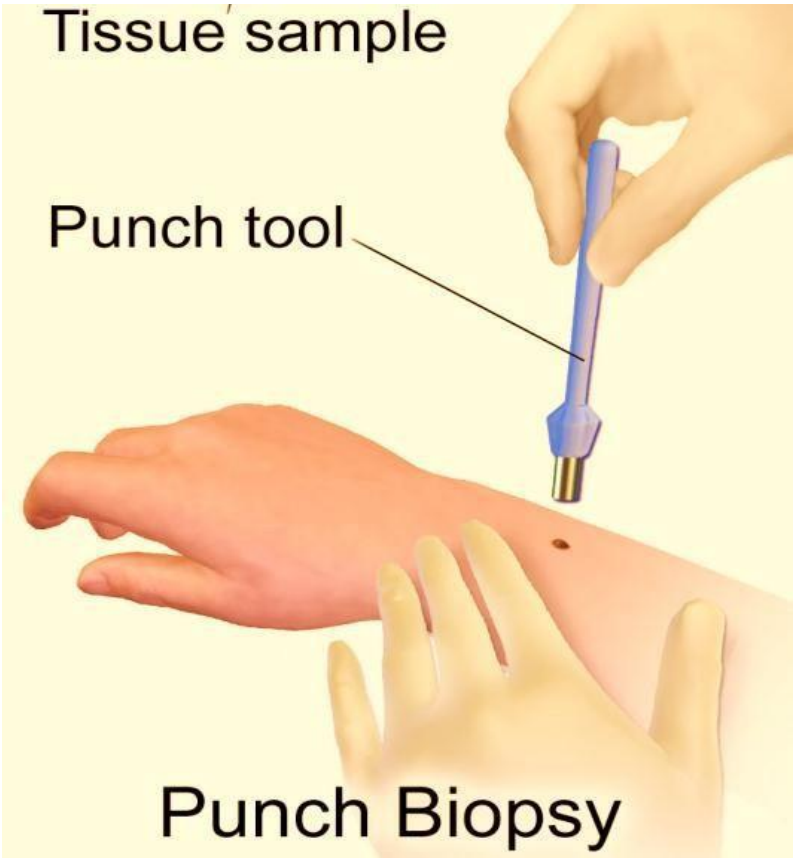
means removal of whole diseased tissue with a rim of normal tissue or organ for histopathological examination.



3. Punch Biopsy:

It is done by biopsy forceps.

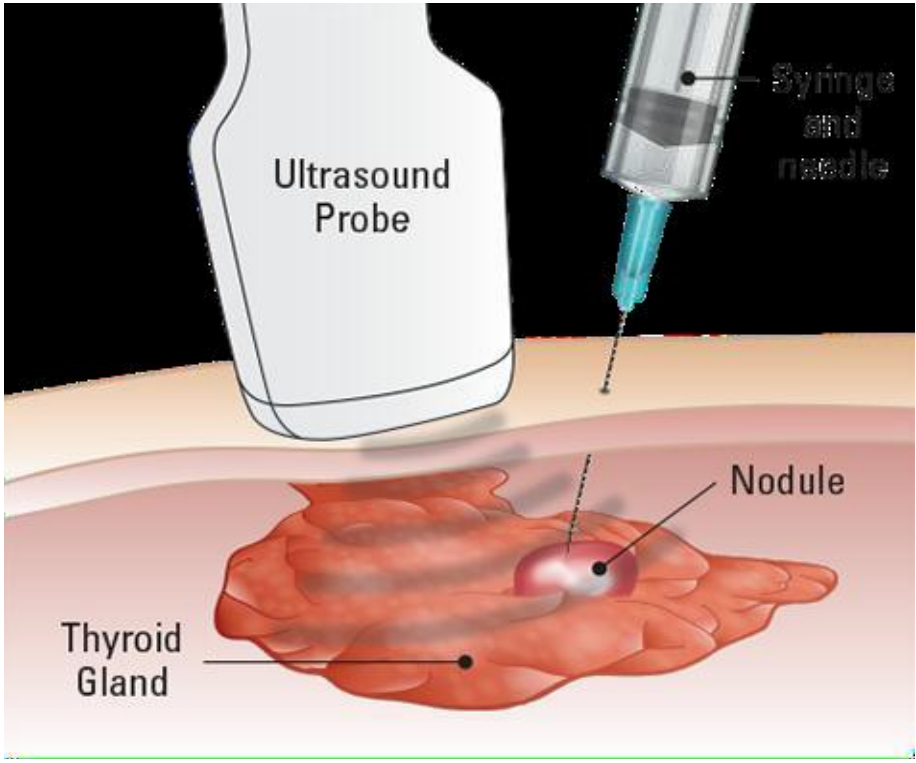
It is performed in the lesion of cervix, oral cavity, esophagus, stomach, intestine and bronchus.



4. Core Needle Biopsy:

It is done with special type of wide bore biopsy needle.

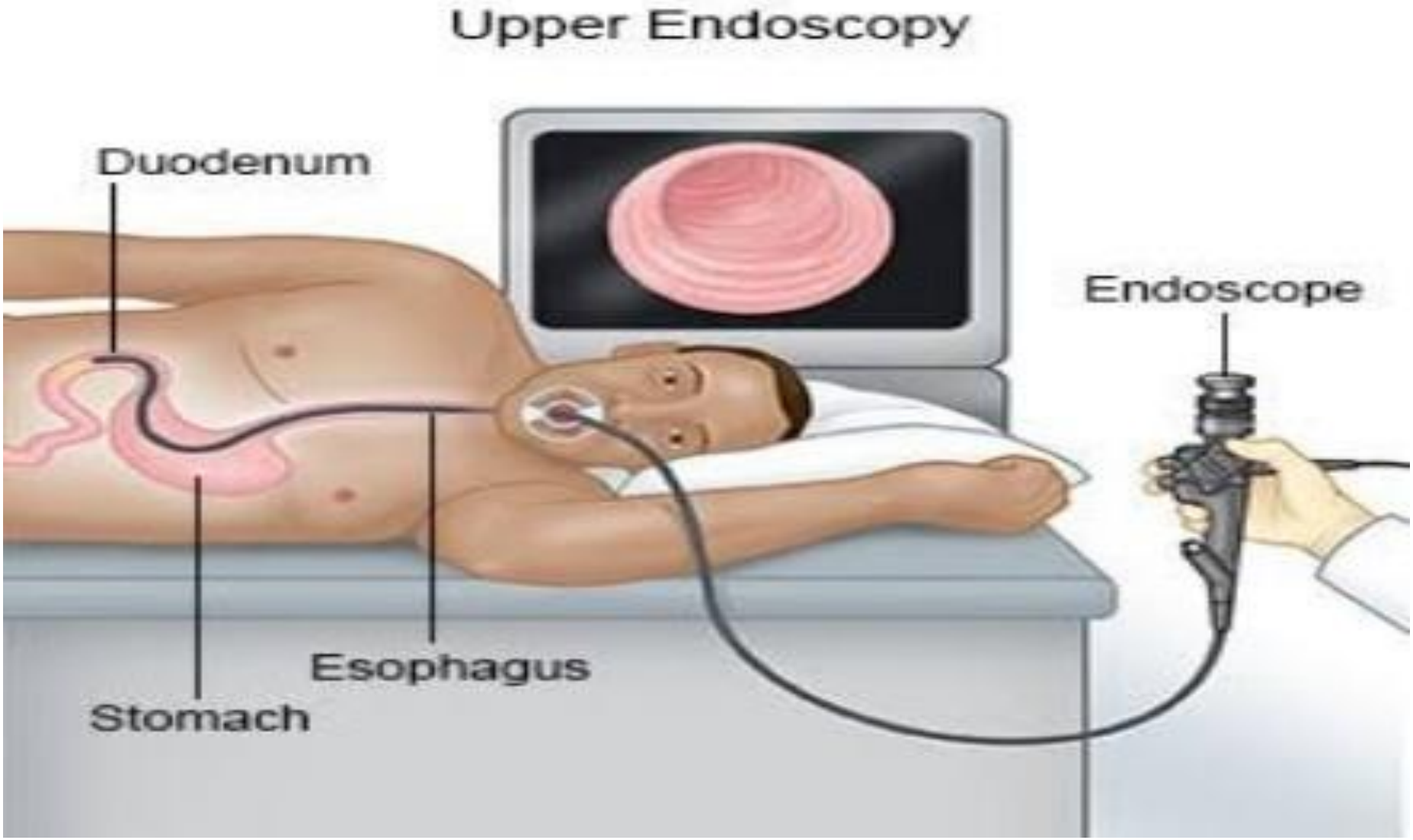
It permits a percutaneous approach to internal structures.



5. Curettage Biopsy:

Curetting are usually done for diagnosis of endometrial disease





6. Endoscopic Biopsy

Laparoscopic Biopsy



❖ **Fine needle aspiration cytology:**

means aspiration of cells from a tissue or organ by a syringe for microscopical examination.

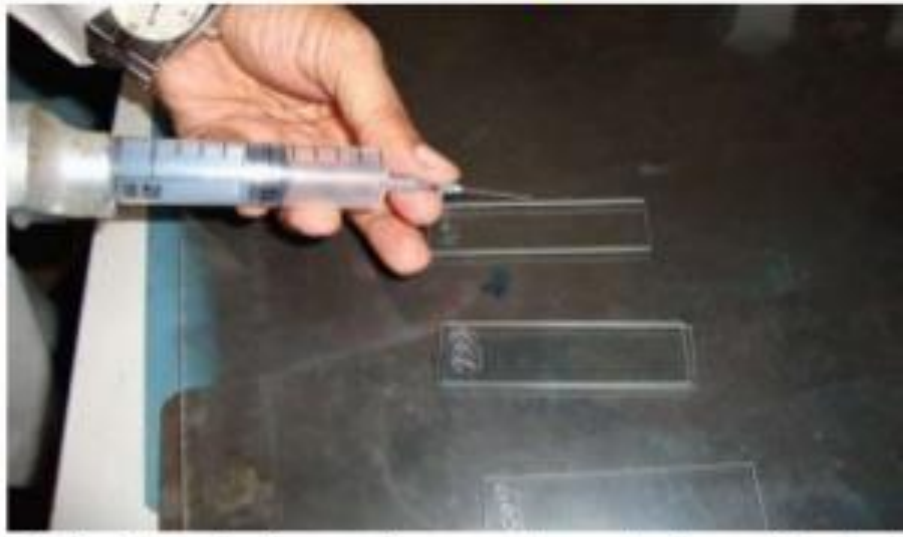
FNAC with aspiration



Site of FNAC should be cleaned by spirit swab



Needle is introduced in the swelling and is gently moving to and fro. Simultaneously negative suction is also created by withdrawing the piston



Air is taken in the syringe and needle is reattached



The aspirated material is expelled and the smear is made by gently pressing the upper slide on the lower one

Sending specimens for pathologic examination:

1. The Request form should be labeled with:
   a) Patient's information.
   b) Hospital/health facility information.
   c) Clinical notes / surgical findings.
   d) Description of the organ , tissue or the site from where the specimen was taken.

2. The Biopsy container should be labeled with the patient name , hospital name, ward number, name of physician and date.
   The specimens should be put in fixative solution immediately after removal the volume of which should be at least 10 times the size of the specimens, 10% Formaldehyde is commonly used.

Department of Pathology

M Abdur Rahim Medical College, Dinajpur.

Histopathology Requisition Form

Reg. No: ..... Date : .....

Patient's name: ..... Age: ..... Sex: .....

Surgeon: .....

Ward: ..... Unit: ..... Bed No:..... Cabin No.....

Name of specimen: .....

Clinical history: .....

X-Ray and other investigations report: .....

Clinical diagnosis: .....

LMP (If specimen is endometrium): .....

Patient's mobile no: .....

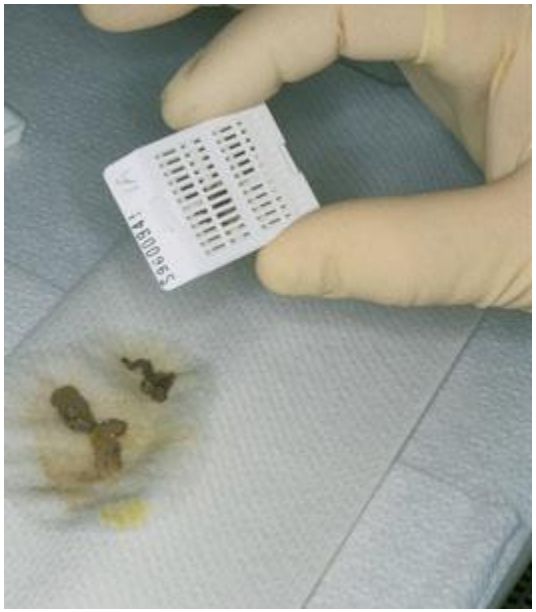
Signature of Doctor

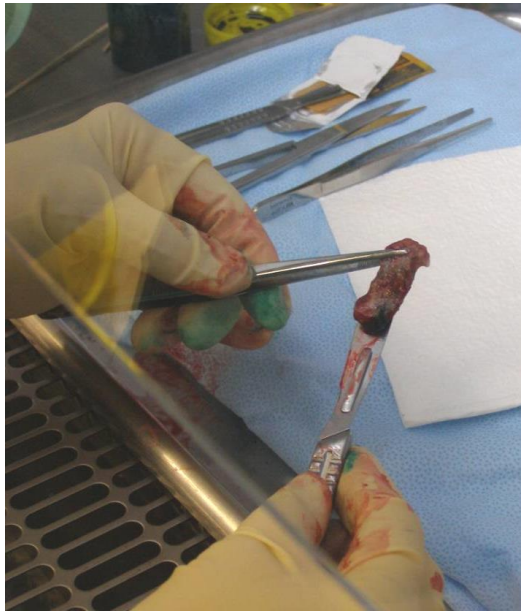

BIOHAZARD

Gross examination:





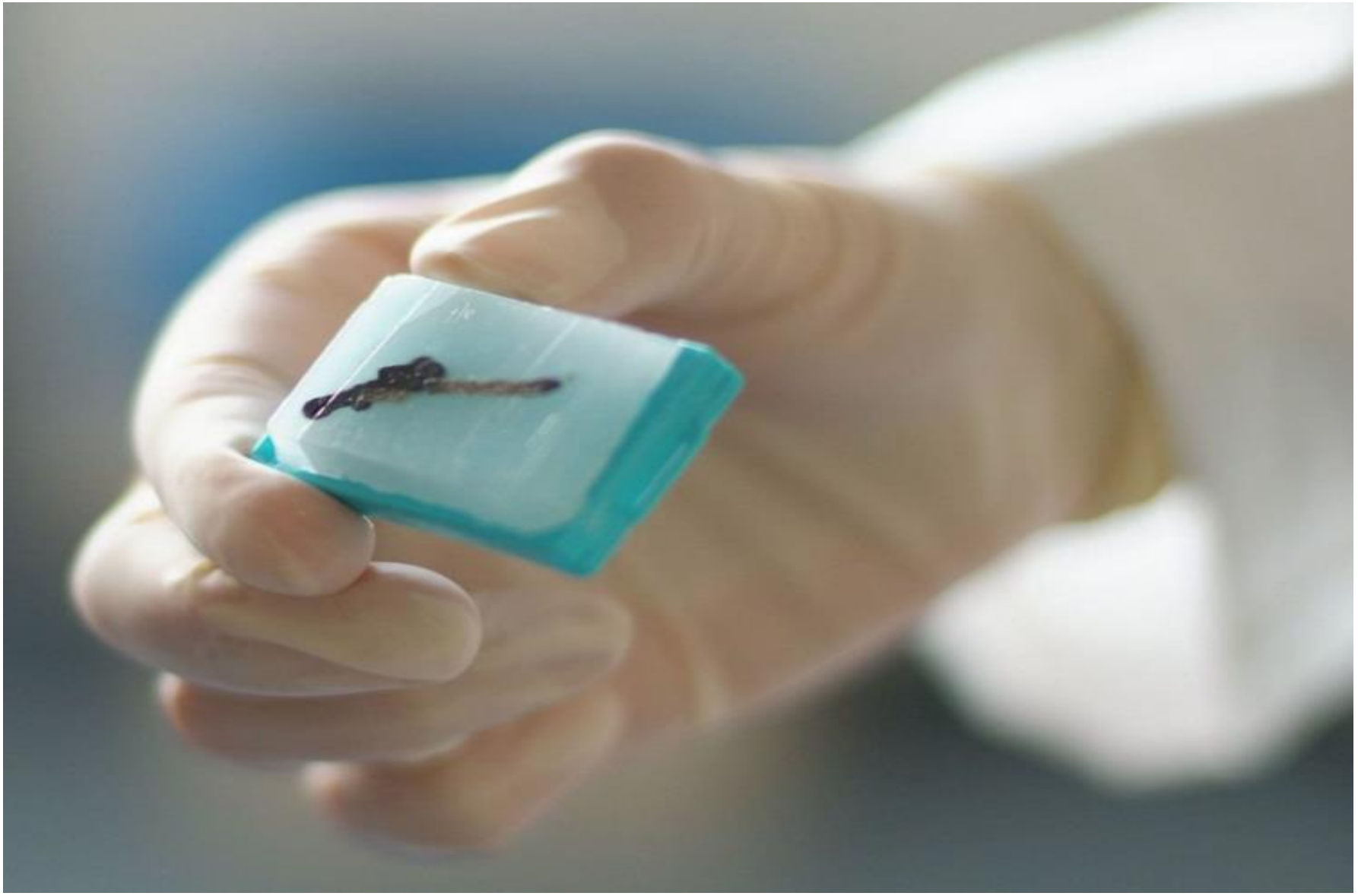




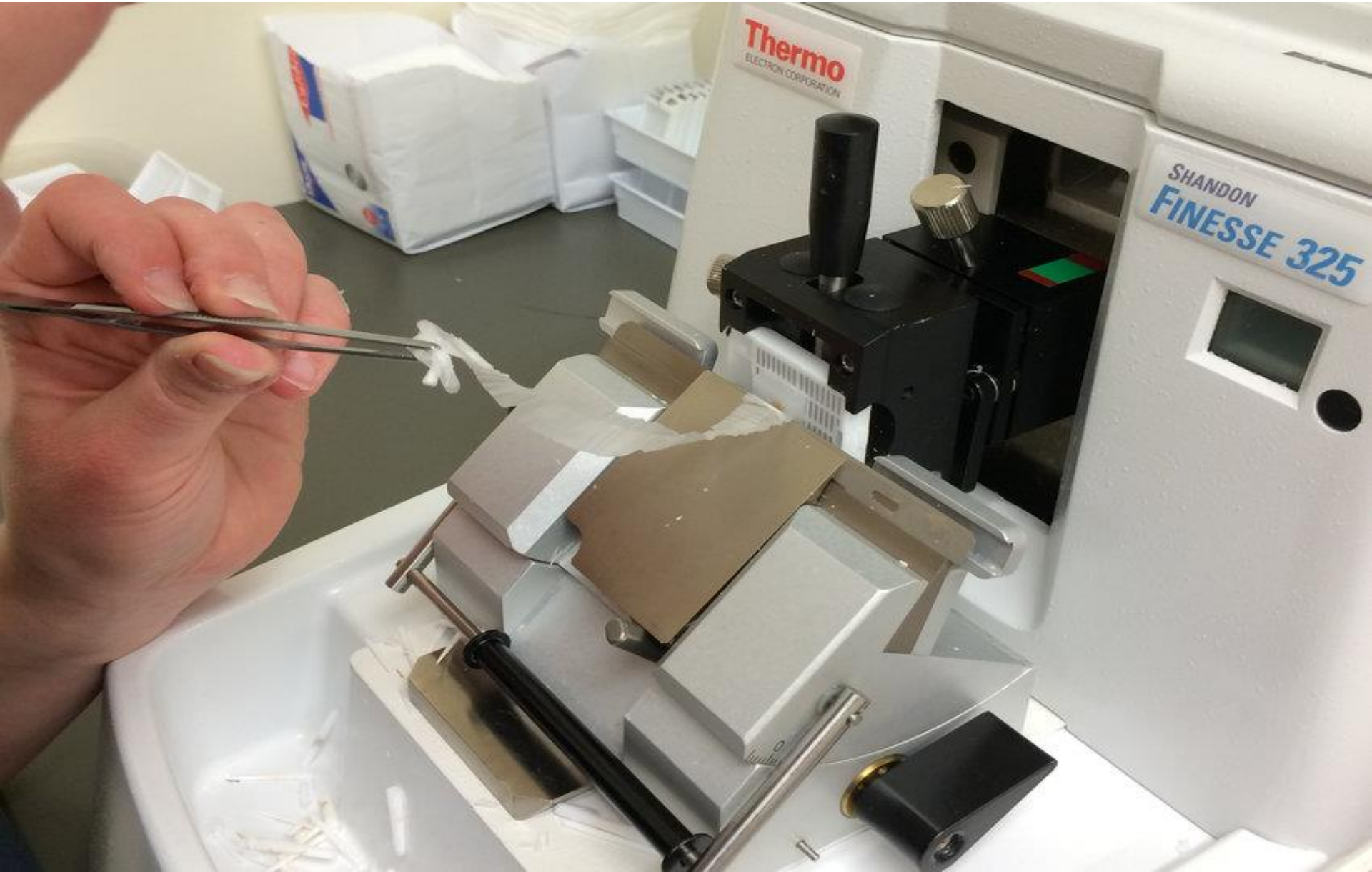
Tissue processing:



Paraffin block.



Sectioning:



Staining:


ABC123 John
Smith, Left chest WA
HE 3
A
ILLU PATHOLOGY
ABC123 John
Smith, Left chest WA
HE 3
A
ILLU PATHOLOGY







Hospital Name
Address

Surgical Pathology Report

Patient: Last Name, First Name
MRN: Medical Record Number
DOB: Date of Birth (Age: #)
Gender: M/F

Accession Number: Specimen Identification
Procedure: Date
Attending: Doctor's Name

Clinical History: Large Gastric Mass

Specimen: Gastric Mucosa

Diagnosis

Stomach, Partial Gastrectomy:

- Malignant Epithelioid Gastrointestinal Stromal Tumor
- Tumor Size 10 x 9 x 8 cm
- Cell Type: Epithelioid and Spindled
- High cellularity; present
- Mucosal Invasion: Focally present adjacent to ulceration
- Mucosal ulceration present
- Mitotic Count: 10/50 HPF
- Myxoid background: Focally present
- Foci of necrosis present
- CD117, vimentin, and CD34: uniformly positive

Gross Description

The specimen consists of an approximately 5 x 7 cm portion of gastric mucosa that is surrounded and underlying by a lobulated mass which is 10 x 9 x 8 cm. The central portion of the mass appears to have an approximately 1.5-cm ulcer. The mucosa away from the area of ulceration is partially removed from the underlying tumor. The underlying mass appears encapsulated and lobular. Gross sections show the lesion to consist of several different patterns. A single area has a gray to gray-tan pattern with an area of central necrosis showing a fairly uniform appearance whereas; other regions of the tumor are gray white- and somewhat lobular in appearance. Areas of yellow necrosis are scattered through the tumor. Representative portions submitted.

Microscopic Description

Sections through the neoplasm show it to be primarily a high cellular neoplasm. The cells are in part arranged in fascicles and clusters with enlarged elongate nuclei having relatively find nucleoli. In some areas, the fascicles have an interwoven appearance. Mitotic figure up to 10:50 HPF. A few areas show foci of necrosis with the cells appearing to be surrounded by somewhat myxoid stroma. Foci of displayed necrosis are present. The lesions appear circumscribed, although not specifically encapsulated. It focally involved the mucosa and shows full thickness ulceration. The tumor immediately beneath the mucosal area of ulceration has a nearly lobular somewhat spindled growth pattern. Some areas of the tumor have a slightly more rounded nuclei and somewhat epithelioid appearance. The cells appear to be arranged in groups and clusters. Some of the cells have cyptoplasmic vacuoles. These areas also show a prominent mitotic activity. Some mitotic figured are abnormal and atypical. The tumor contains numerous relatively open vascular channels which appear to be part of the neoplasm. The tumor has a pseudo capsule and in some areas appear to be nearly covered.

Immunostains are strongly positive for CD117 (C-kit), CD34, and Vimentin, Smooth muscle actin, Desmin, Synaptophysin, S-100, and Ck8/18 are negative.

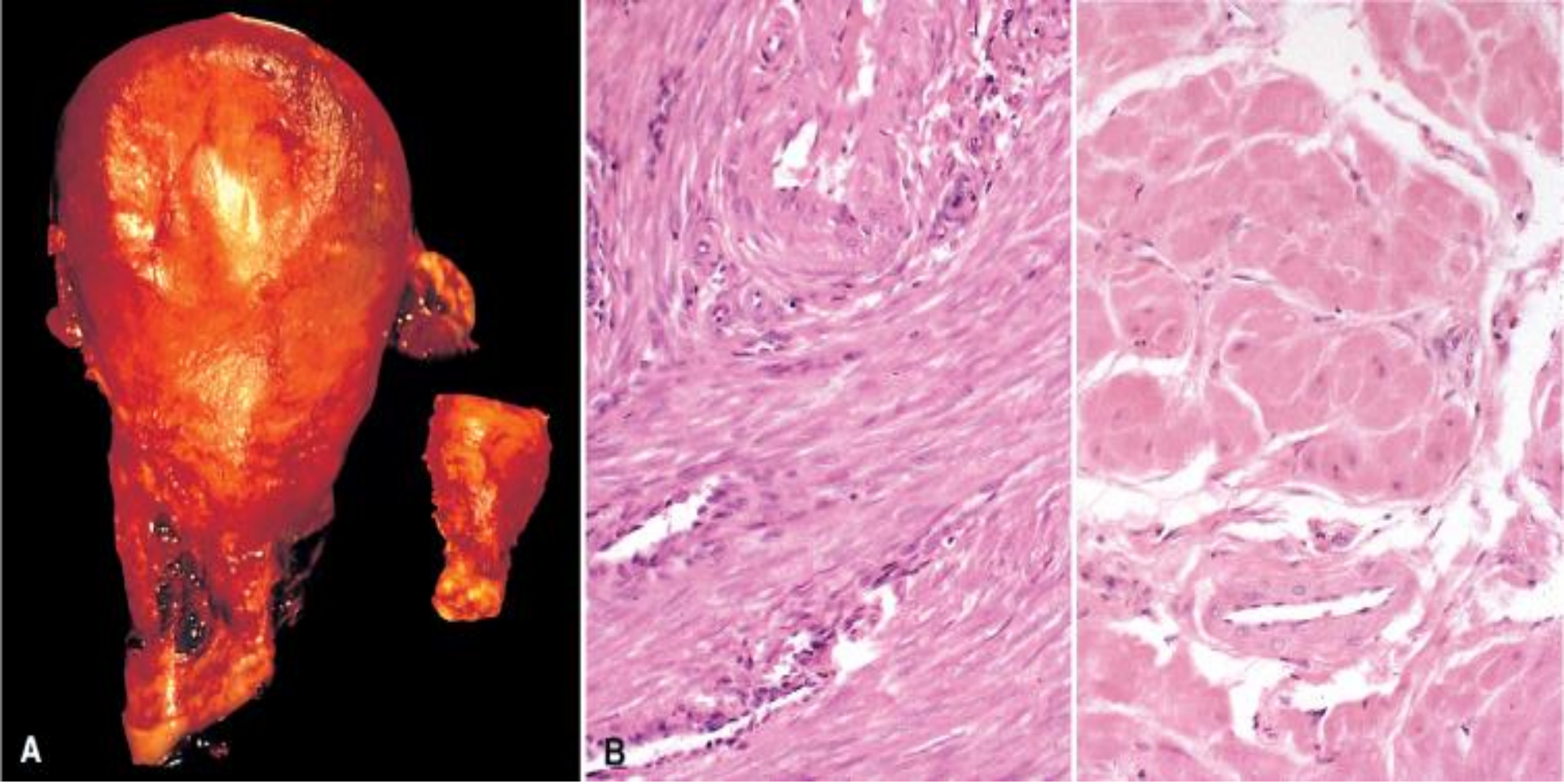
Comment

Immunostains were performed on the core biopsy and demonstrate that the tumor cells are positive for CD117. The findings are consistent with the above diagnosis.

Cell injury
Lab. 2



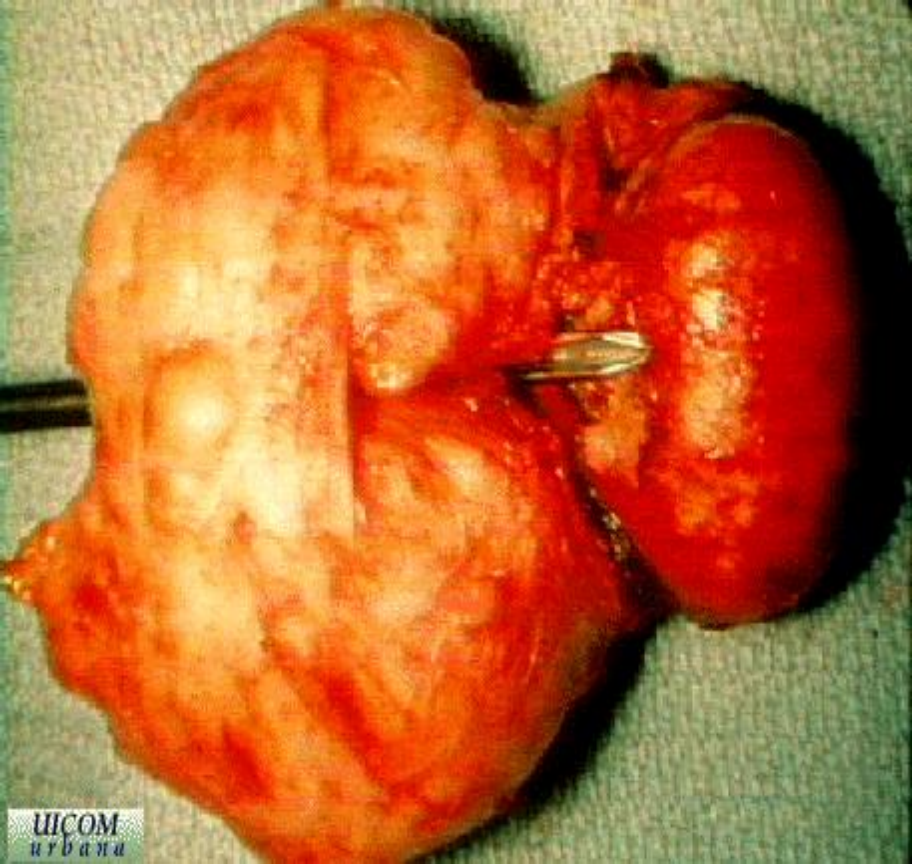
Left ventricular hypertrophy in patient with chronic hypertension

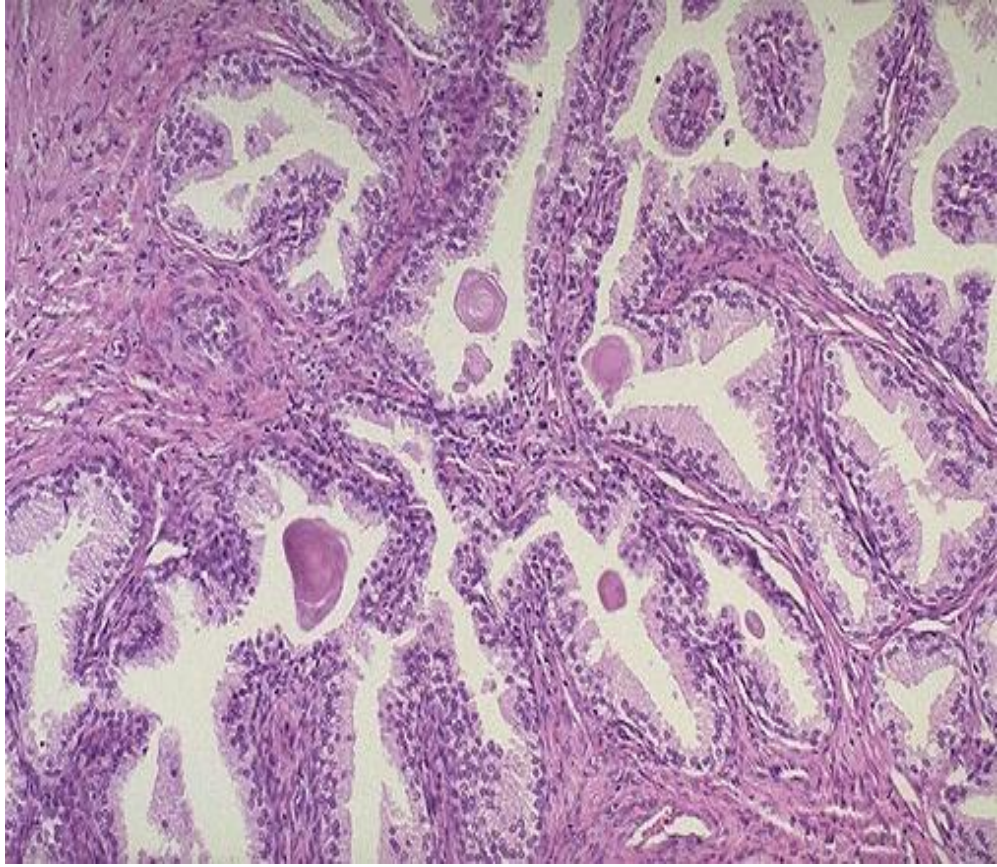


Gravid uterus(physiologic hypertrophy & hyperplasia).

A: Gross appearance.

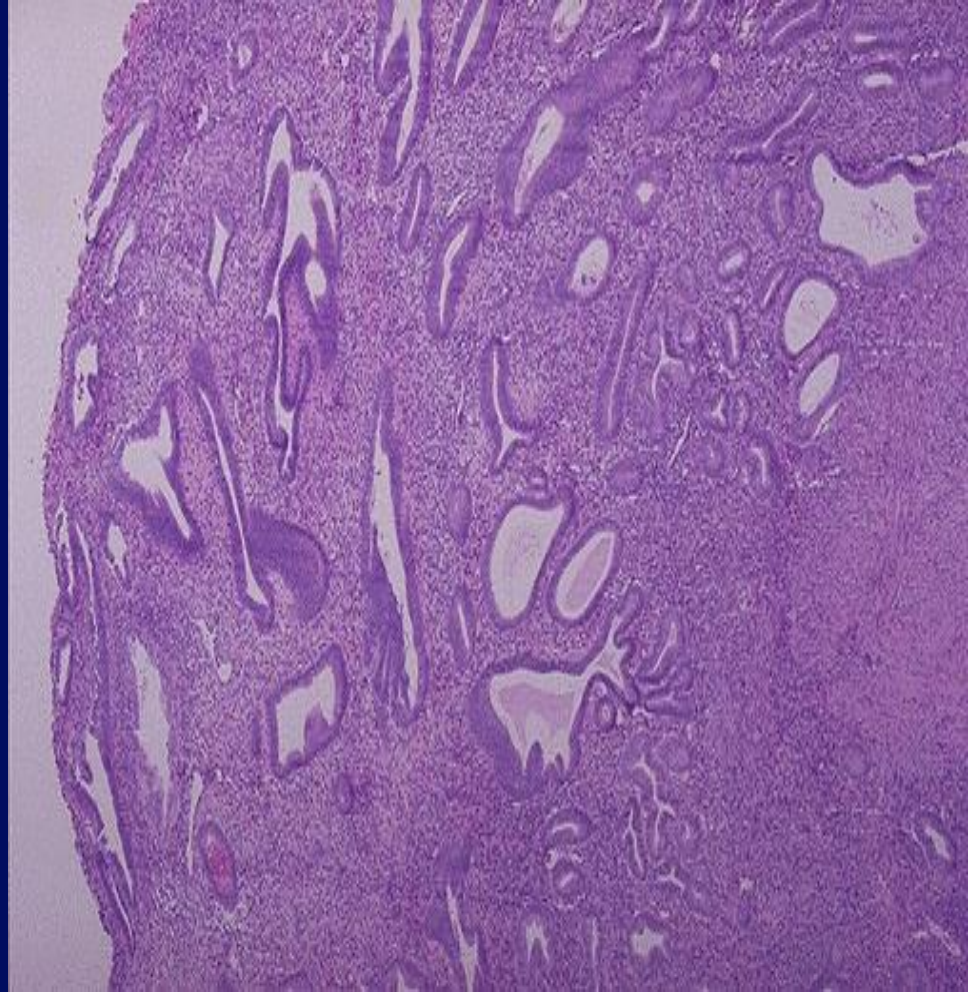
B: Small spindle-shaped uterine normal smooth muscle cells (left) compared with large plump cells in gravid uterus (right)





Benign prostatic hyperplasia (hypertrophy)
Pathologic hormonal hyperplasia





Endometrial hyperplasia

(Pathologic hormonal stimulation) (estrogen)



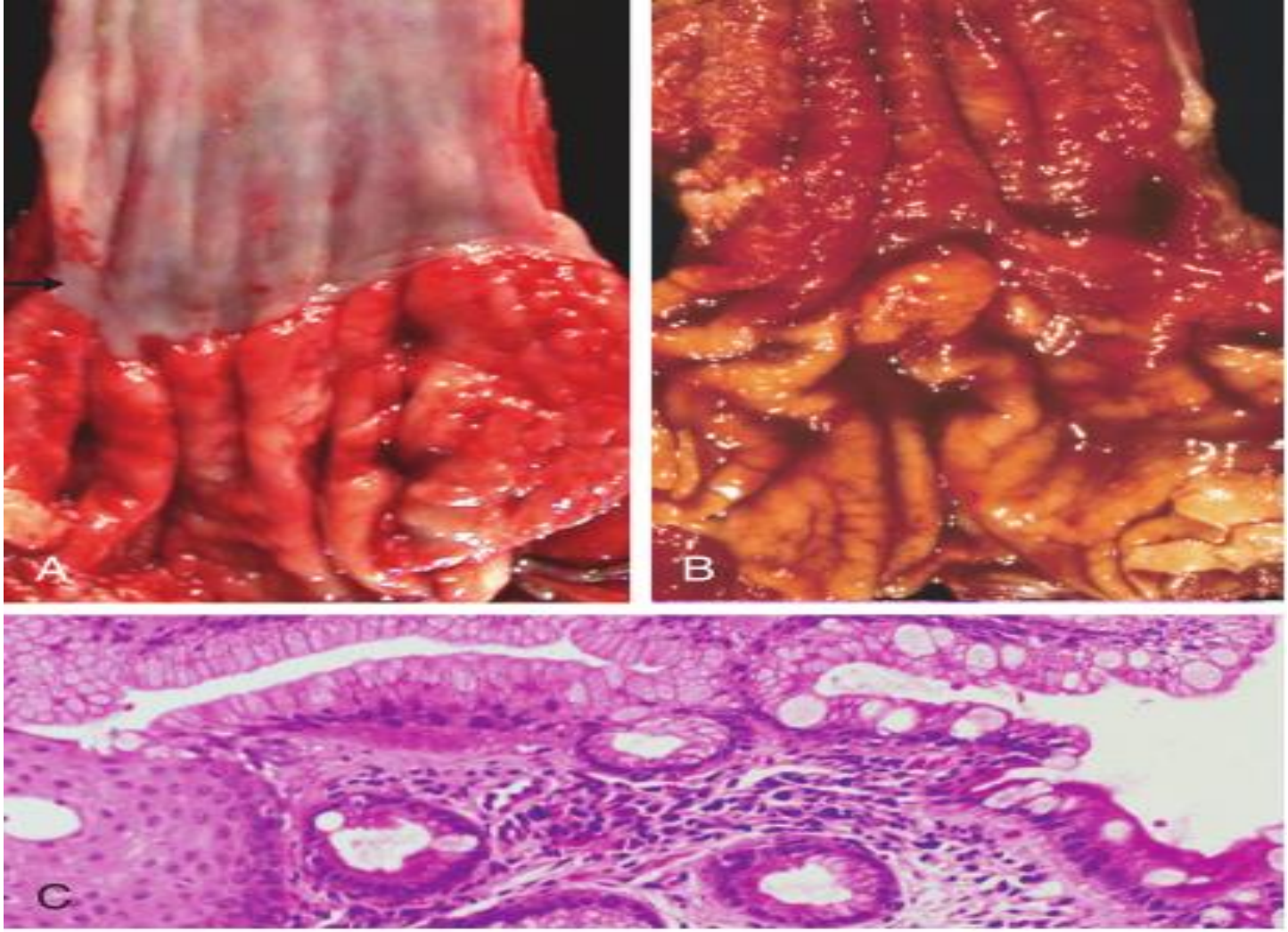
Atrophy as seen in the brain. Left: Normal brain of a young adult. Right: Atrophy of the brain in an 82-year-old man with atherosclerotic disease. Atrophy of the brain is caused by aging and reduced blood supply. Note that loss of brain substance narrows the gyri and widens the sulci

Disuse atrophy



Testicular atrophy

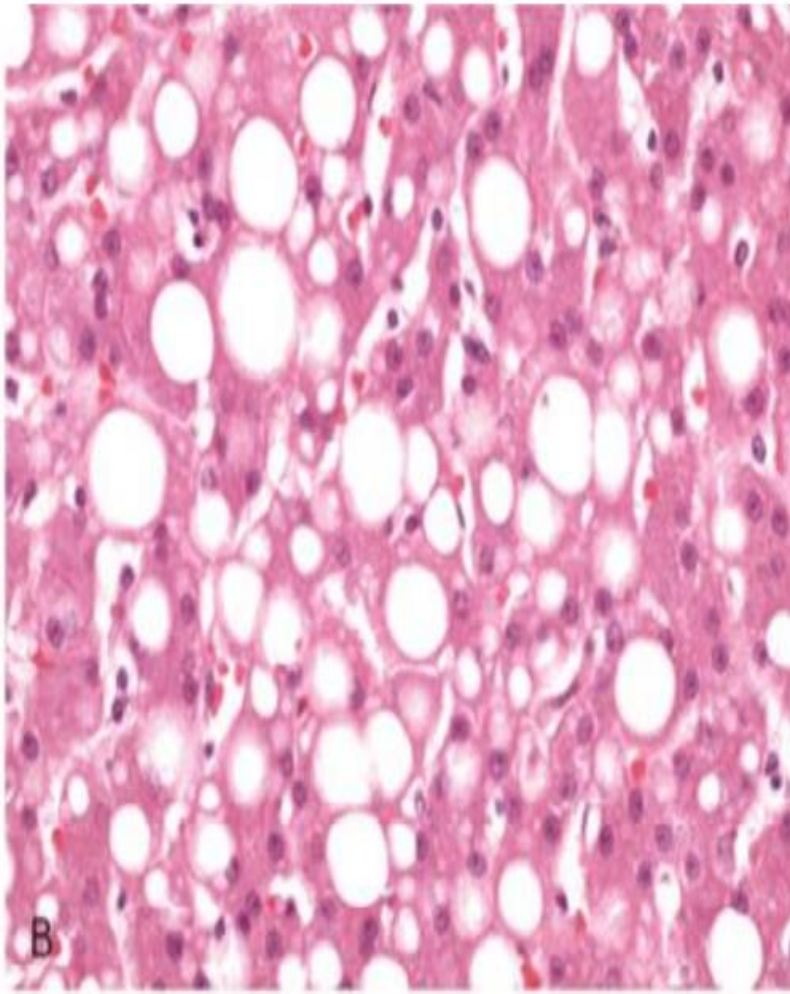




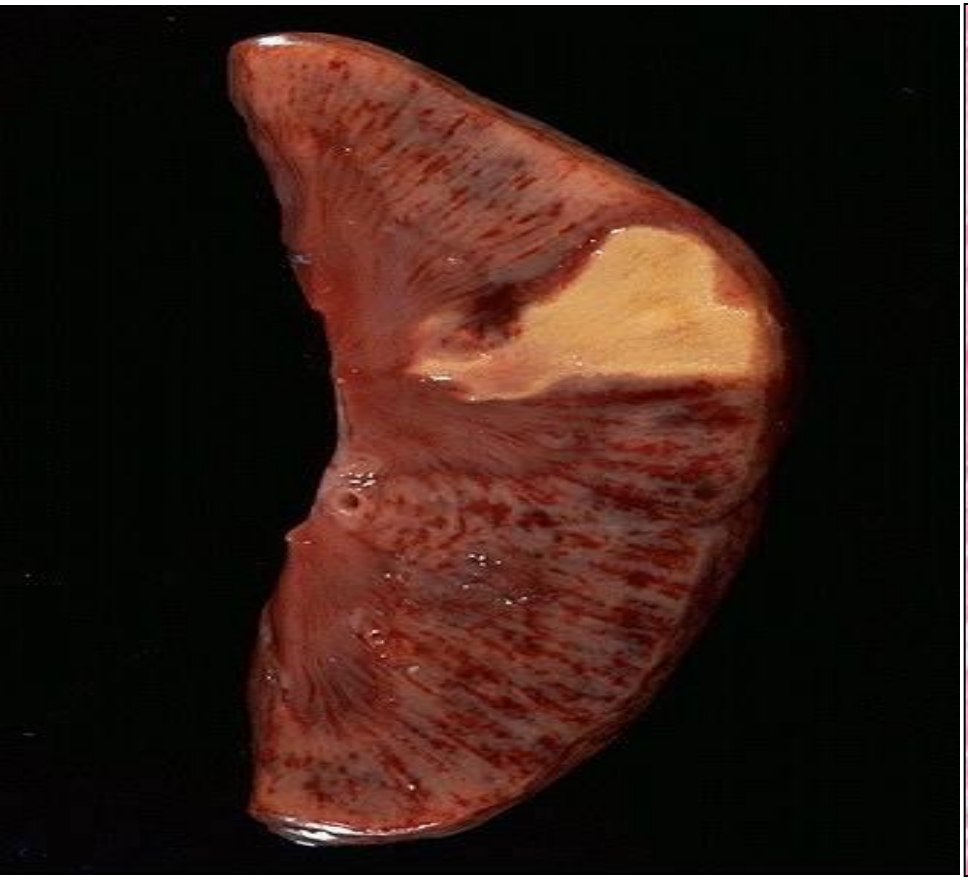
Barrett esophagus. A, Normal gastroesophageal junction B, Barrett esophagus. Note the small islands of paler squamous mucosa within the Barrett mucosa. C, Histologic appearance of the gastroesophageal junction in Barrett esophagus (squamous to columnar).

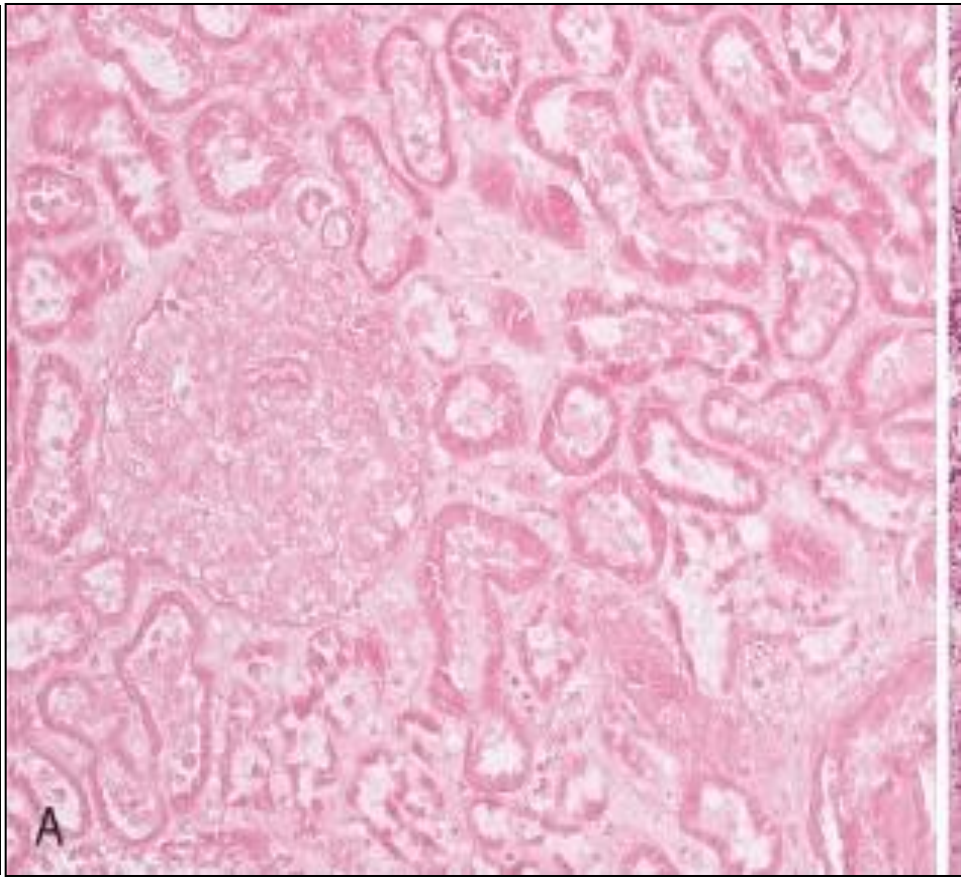
Fatty change- liver





Coagulative necrosis



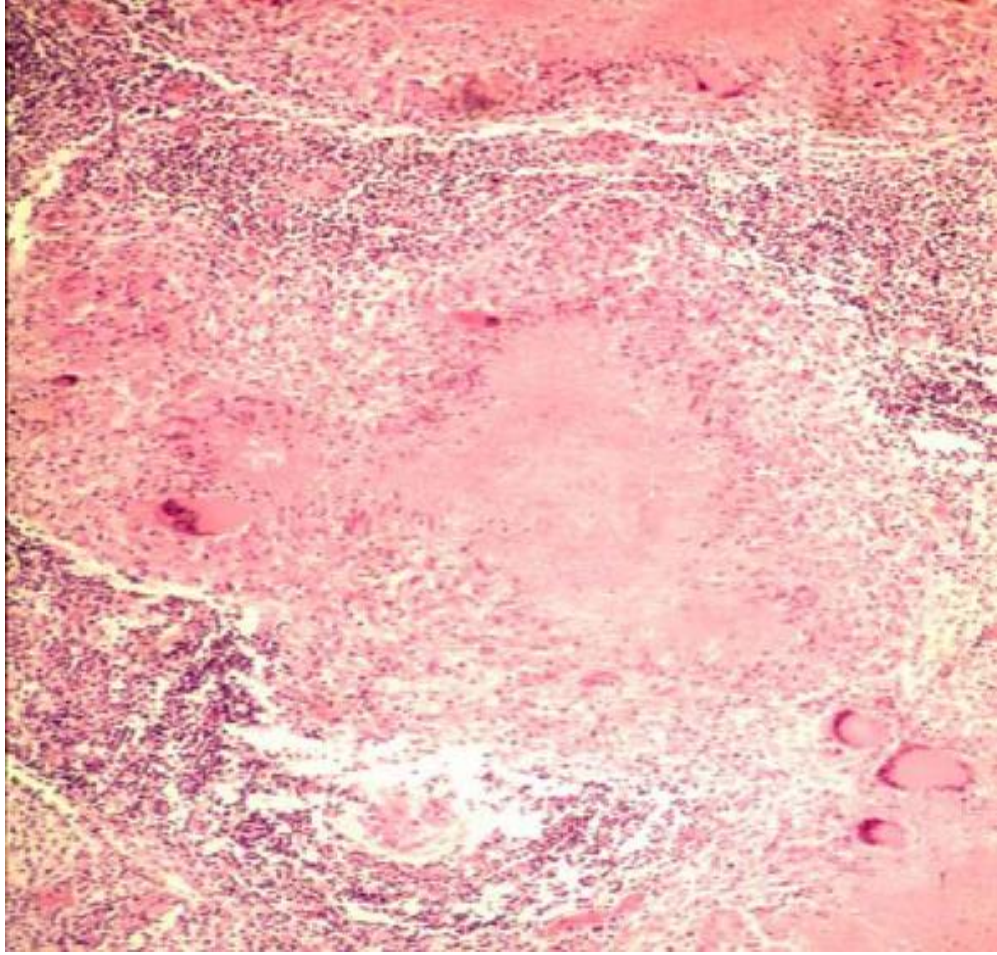


Kidney infarct exhibiting coagulative necrosis, with loss of nuclei and clumping of cytoplasm but with preservation of basic outlines of glomerular and tubular architecture

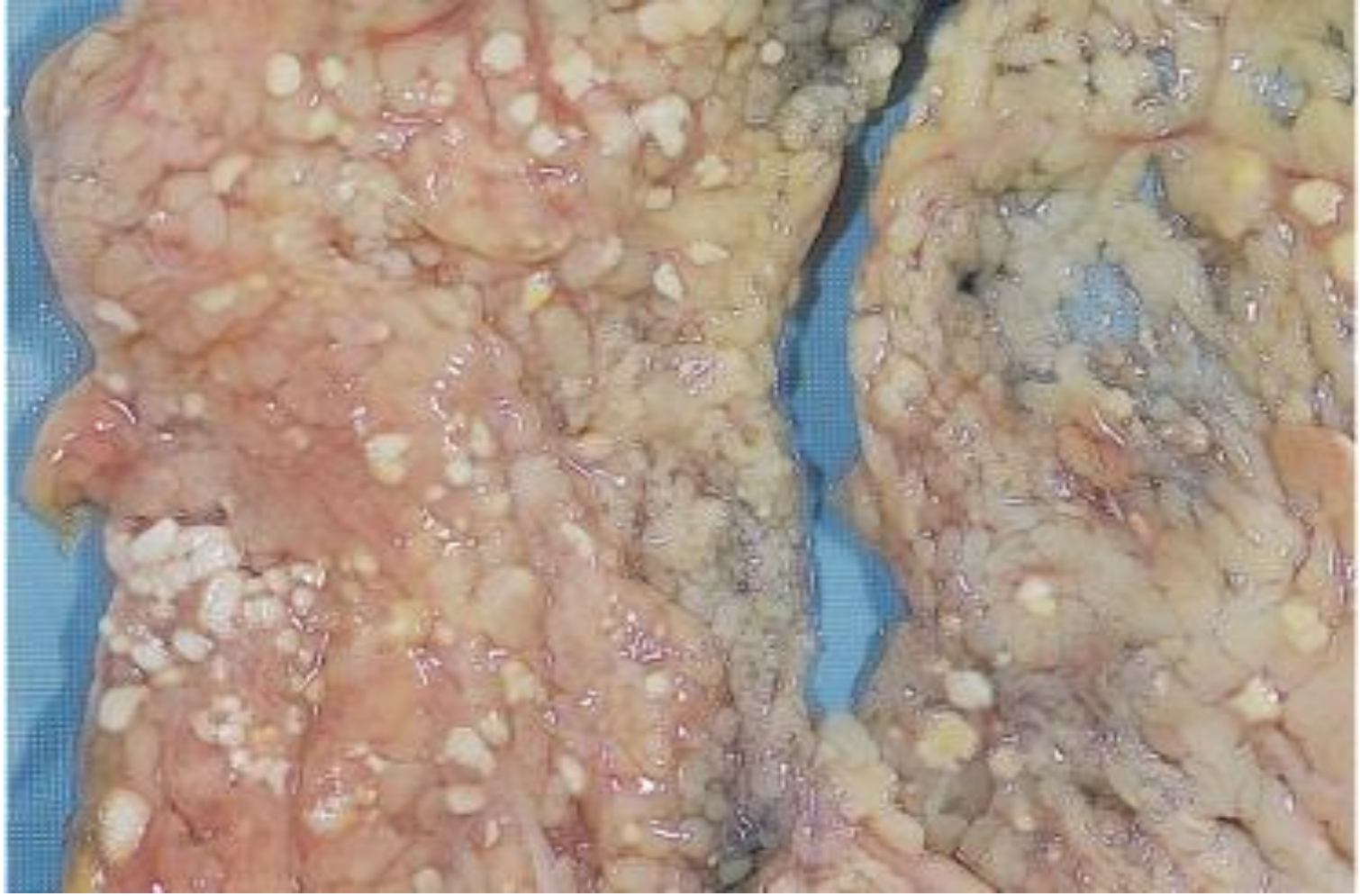


Liquefactive Necrosis: Brain



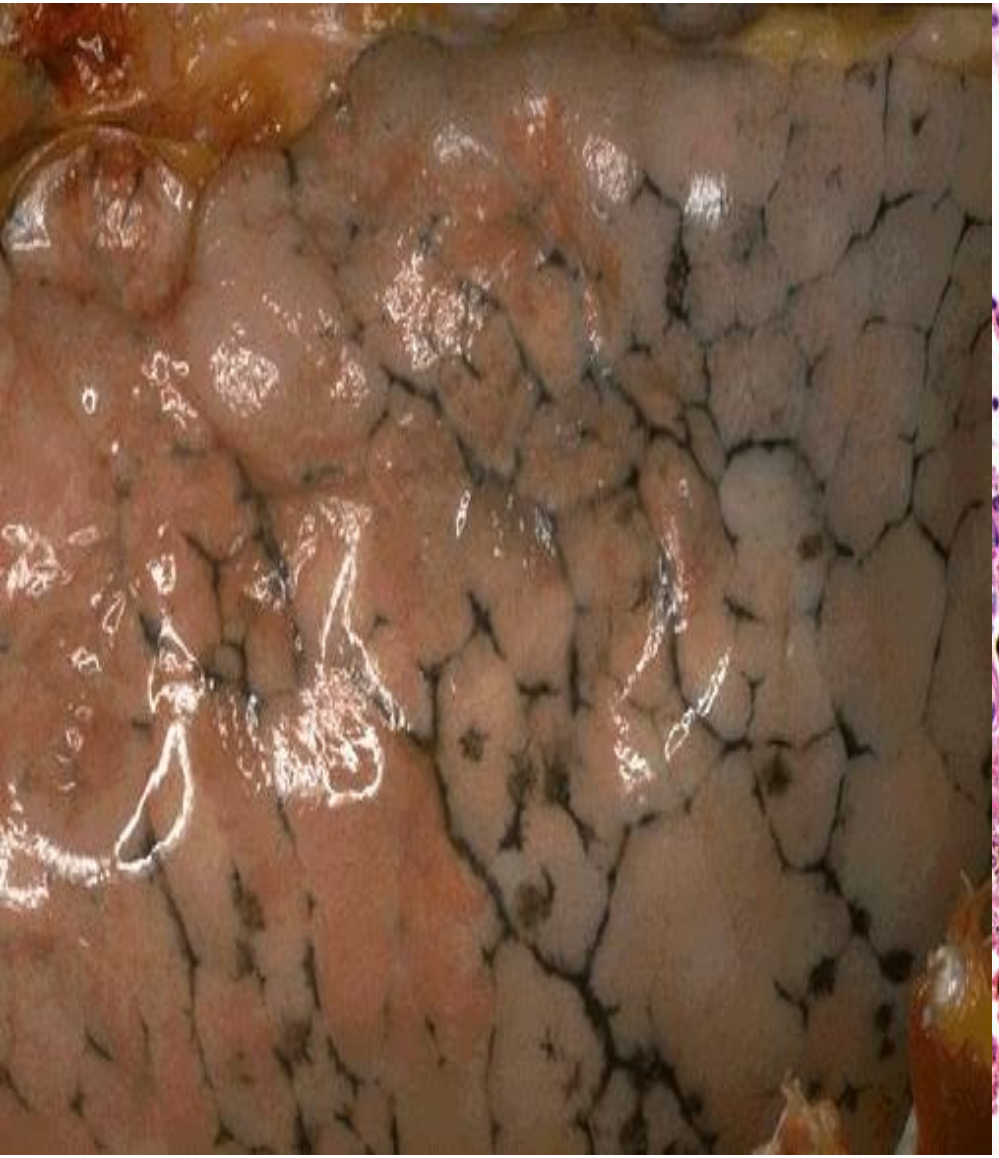


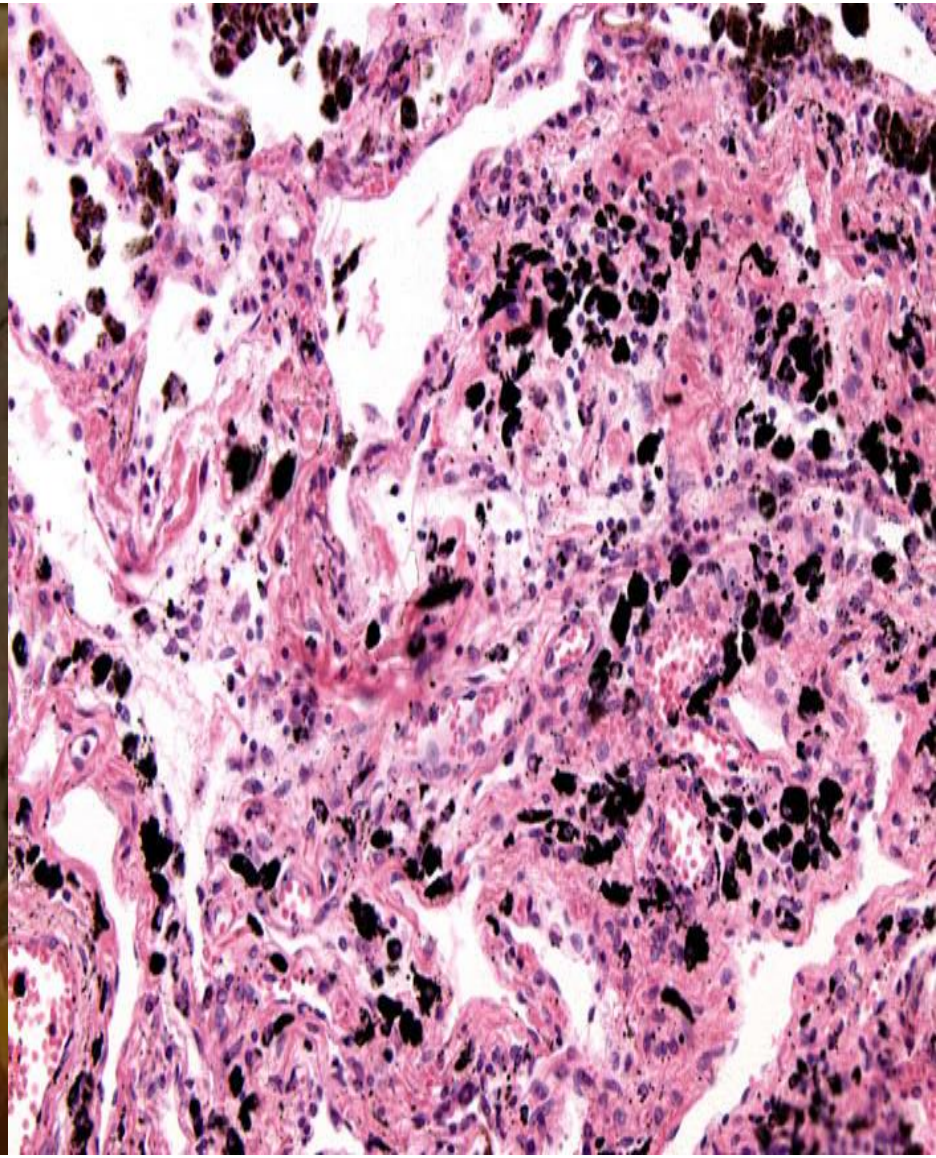
Caseous necrosis: Lung



Foci of Fat necrosis with saponification in the mesentry. The areas of white chalky deposits represent calcium soap formation at sites of lipid breakdown

Anthracosis - lung







Aortic valve , gross ,
(calcified aortic stenosis)

End